Nikol Savova

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Education

University of Oxford, Wadham College — Oxford, UK
MMath Mathematics (integrated master's) — *Expected 2027*

- Coursework included Probability, Measure & Martingales / Statistics / Graph Theory / Galois Theory / Algebraic Number Theory / Logic / Set Theory / Complex Analysis / Topology / ODEs & PDEs.
- Mini-dissertation on Fermat's Last Theorem and the attempts preceding the proof (History of Mathematics), awarded 80/100.

Sofia High School of Mathematics — Sofia, Bulgaria
Specialist mathematics school, olympiad track. GPA 6.0/6.0 — *2023*

Awards & Competitions

- Emergent Ventures Fellow, Mercatus Center, George Mason University (2023), grant awarded by Tyler Cowen.
- Bronze Medal, Bulgarian National Winter Mathematics Competition (2021).
- Bronze Medal, Bulgarian National Mathematics Tournament (2019).
- Selected to represent Bulgaria, International Young Naturalists' Tournament, physics & mathematics (2019).

Research

Proof Hunter
Collaborative research on Brenti's log-concavity conjecture for Bruhat intervals — *2026–present*

- Substantially extended the computational verification of the conjecture beyond prior published results, and co-authoring a paper on it, now nearing submission.

Einstein Institute of Mathematics — Sofia, Bulgaria
Research Assistant to Dr. Igor Balla, spectral graph theory — *May–Oct 2023*

- Worked on generalising the Alon–Boppana theorem from regular to arbitrary graphs, replacing the adjacency matrix with the graph Laplacian and taking the Laplacian spectral gap (Fiedler's algebraic connectivity λ_2) as the object of interest.
- Sought lower bounds on the spectral gap that remain meaningful in the dense regime, where the classical Alon–Boppana bound degenerates as the diameter collapses, and related these bounds to the positive discrepancy $\mathrm{disc}^+(G)$.
- Implemented computational searches in Python to test conjectures and construct candidate examples.

Experience

AI Alignment Research Internship
Incoming (mathematical and ML foundations of AI alignment) — *2026*

Mercor
Generalist Expert (Contract) — *Aug 2026–Present*

- Evaluate and provide structured feedback on AI-generated outputs for AI research partners.

Bloomsbury Technology — London, UK
Head of Quantitative Research (pre-seed) — *Jan–Apr 2026*

- Early quantitative research for a causal and behavioural investment engine for the art market, building data-scraping pipelines and leading feature selection for valuation datasets.
- Ran market discovery with financial-consulting professionals to identify the company's initial target segment. Studied statistical machine learning throughout.

Projects

Proof Hunter (AI-assisted mathematics)
github.com/NikolSavova/proof_hunter — *2026–present*

- Built a pipeline for discovering open mathematics problems that are tractable, verifiable, and genuinely unsolved. It ingests problems at scale from curated sources, scores them via LLM triage, checks openness against the published literature, and produces a ranked shortlist.

Programs & Teaching

PAIR 2026 (FABRIC)
Junior Counsellor (staff) — *2026*

- Selective international summer program on rationality and AI reasoning ($\sim$30 students/cohort worldwide).

ESPR 2023
Junior Counsellor (staff) — *2023*

- Also attended WARP 2022, ESPR 2022, and Future Forum 2022.

Easymath, Sofia
Mathematics Instructor — *2022–2024*

- Designed olympiad curricula and original problem sets, taught weekly classes for 70+ students.

Skills

Programming Python, C++, Lean 4 (in progress), Git, LaTeX
Mathematics probability theory, stochastic processes & martingales, measure theory, statistical inference, spectral graph theory, extremal combinatorics, linear algebra, real & complex analysis
Certifications Supervised Machine Learning Specialisation, DeepLearning.AI / Stanford (2025)
Languages Bulgarian (native), English (fluent), German (working proficiency)